

This design turns one ordinary UniFi LAN into several small trust zones without renumbering everything at once. It favours memorable names and addresses, but keeps the binary subnet boundaries that routers require. Every change has a proof step and a retreat path. Private instance values—people, device identifiers, keys and exact assignments—belong in the private instance repository, not in this publication.

Rules that do not move

One site root	The first site uses a private 10.1 root. This is an allocation convention, not one enormous scannable subnet.
Small active ranges	Each zone receives the smallest power-of-two subnet with between roughly two and ten times its expected population. Growth is deliberate; scans target one CIDR at a time.
Human first	Gateways normally end in .1; durable services take the next simple addresses; people use DNS names rather than memorising numbers.
One DHCP authority	UniFi assigns addresses. A boot server supplies files, not competing leases. PXE information is attached only to the provisioning network.
Deny between zones	A new zone may reach nothing until its required flows are named. Internet access, DNS, identity, storage and administration are separate permissions.
Public pattern, private instance	This document contains roles and placeholders. The private inventory contains usernames, hostnames, serials, MAC addresses, reservations, share names and actual CIDRs. Secrets remain encrypted even there.

Addressing model

Subnetting is binary even when the visible plan uses decimal-friendly labels. A ten-address subnet is not a valid routing boundary; a /28 provides 14 usable addresses. The operator sees the gateway at the first usable address and a short, documented range. Unused addresses are not advertised as another network.

Expected now	Initial CIDR	Usable addresses	Expansion decision
1–3	/29	6	Use only for genuinely fixed, tiny roles; reserve an adjacent block.
4–7	/28	14	Default for a small appliance class.
8–15	/27	30	Default for workstations or a growing device class.
16–30	/26	62	Use only when the inventory already justifies it.

Capacity rule
Review a zone at 70 percent occupancy. Expansion is a planned subnet change, not a reason to begin with a /16 or scan an entire private address block. Reserve nearby address space in the plan, but create only the network in use.

Zone register

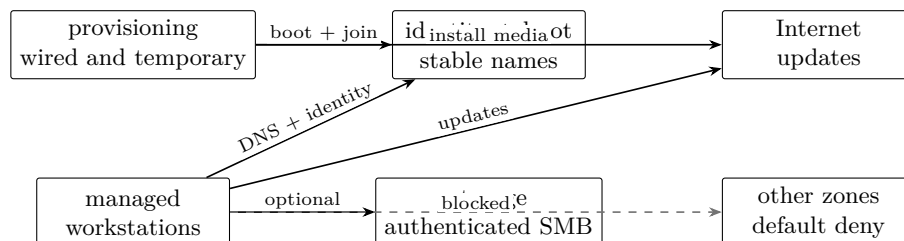
Zone numbers should be memorable decimal labels, commonly in tens. They do not have to equal an IP octet or encode security policy. The private instance file binds each placeholder to an exact VLAN and CIDR.

Zone role	VLAN holder	place-	Initial size	Purpose
Management	VLAN_MGMT		/28	UniFi consoles, switches and access points; administrator devices only.
Identity and boot	VLAN_CORE		/28	AD DNS, Kerberos, directory and boot publication services.
Storage	VLAN_STORAGE		/28	Primary and backup storage appliances; no ordinary client membership.
Provisioning	VLAN_BUILD		/28	Wired, short-lived workstation installation boundary.
Managed workstations	VLAN_WORKSTATION		/27	Installed Windows and Arch clients.
Trusted personal	VLAN_TRUSTED		/27	Personal clients not yet under workstation management.
Printers and media	VLAN_DEVICE		/28	Devices needing narrow, explicit inbound access.
IoT	VLAN_IOT		/27	Untrusted household appliances; no lateral access.
Cameras	VLAN_CAMERA		/28	Cameras and recorder flows only.
Guests	VLAN_GUEST		/27	Internet only, client isolation enabled.

Host-address convention

Within a zone, .1 is the gateway. Prefer .2 through .9 for durable infrastructure, keeping fixed addresses outside the dynamic pool. Use DHCP reservations by DNS name for ordinary devices. A trailing zero appears only in unavoidable CIDR notation, never as a host address someone must type. For example, a zone labelled 40 may occupy only <zone-cidr>; the label does not justify routing or scanning its whole parent allocation.

Traffic contract



The matrix below is the source of truth. “Required” still means the rule is limited by source, destination and service; it never means *allow any*.

Source	Destination	Policy	Proof
Managed workstation	AD DNS and identity aliases	Required named services	DNS SRV query, clock check, Kerberos ticket and directory login.
Managed workstation	Storage alias	Optional authenticated SMB	Mapping succeeds when present; login still succeeds when storage is blocked.
Managed workstation	Internet	Required update traffic	Windows and Arch metadata fetch through ordinary DNS; no access to management.
Provisioning client	Boot alias and install sources	Required during a build	DHCP lease, boot filename, iPXE fetch and artifact checksum recorded.
Administrator device	Management zone	Required from explicit sources	Console opens from the allowed source and fails from a workstation.
Guest, IoT, camera	Private zones	Denied unless a narrow exception exists	Attempted DNS, HTTPS and SMB probes fail; rejection appears in logs.
Any client	Peer in same restricted zone	Denied where isolation is promised	Two sacrificial clients cannot exchange test traffic.

UniFi change sequence

Menu labels can move between Network application releases. Match the property named here, not a remembered screen position. Export a UniFi backup and record the Network application version before changing anything.

- Create the zone's virtual network with its recorded VLAN ID, gateway and small CIDR. Keep the UDM as DHCP server. **What this does:** creates one routed broadcast boundary without introducing a second address authority.
- Set a DHCP pool that excludes the gateway and every fixed reservation. Set lease duration deliberately. **Verify:** a test client receives an in-range address, the UDM gateway and the intended DNS servers.
- Assign a sacrificial switch port as an access port for that network. Do not convert a live host link into a trunk during this proof. **Verify:** the port's native network and the client's observed subnet agree.
- Add required allow rules from the traffic contract, most specific first. Add the zone-to-zone deny after each required flow has passed. **What this does:** makes a missing dependency visible instead of hiding it behind broad private-network access.
- Enable logging temporarily on the final deny. Test one allowed and one forbidden flow from the sacrificial client. Remove noisy logging only after the rule counters and client observations agree.
- For the provisioning network only, set the DHCP boot server to the stable boot-service address and the boot filename to the approved x86-64 UEFI iPXE loader. **Verify:** capture the DHCP exchange and confirm that UniFi is the only DHCP OFFER source.
- After ordinary DHCP is proven, enable DHCP Guarding on supported UniFi switches and trust only that network's UDM gateway address. **What this does:** rejects rogue DHCP offers and intentionally excludes Controller ProxyDHCP. **Verify:** a normal UDM lease still succeeds; use an isolated test before generating any deliberate rogue offer.

Pre-attachment gate

Before a port or SSID can carry the new zone, confirm all six: [] backup and version recorded; [] CIDR capacity calculated; [] DHCP pool excludes fixed addresses; [] endpoint and service groups exist; [] final deny is logged; [] rollback port profile is recorded. The operator initials the completed private worksheet; an unchecked item stops attachment.

Version-sensitive console map

Use the search box when a label differs. The object and resulting property are authoritative; the breadcrumb is only a starting point.

Object	Current zone-based releases	Older or alternate label
Virtual network	Settings → Networks → New Virtual Network	Settings → Networks → Create New Network
Zones	Settings → Security → Zones	Implicit LAN, Guest, WAN and Gateway categories
Traffic rule	Settings → Security → Policy Engine	Settings → Firewall & Security → LAN In / Guest In
DHCP boot values	Network → DHCP Service Management → Network Boot	Network → Advanced → DHCP Options / TFTP Server
Port membership	Devices → switch → Ports → selected port	Port Manager or a named Switch Port Profile
DHCP Guarding	Network or switch security settings	DHCP Snooping / Guarding under the selected network
PPSK	WiFi → SSID → Security → Private Pre-Shared Keys	WiFi → Manual / Advanced security

Do not translate rule direction by name

On a zone-based release, express source zone, destination zone and service directly. On a legacy release, an inter-VLAN packet is commonly filtered on the source network's "In" chain. Confirm with a sacrificial packet and the matched-rule counter; never infer equivalence from a similar-looking label.

Staged zone procedure

Create policy before attaching a real client. This avoids the interval in which a newly routed network inherits broader access than intended.

1. Freeze a worksheet row: Network application version, backup identifier, zone name, VLAN, CIDR, gateway, DHCP range, DNS, NTP and expected population.
2. Create address groups for the exact Controller, identity/DNS, storage and administrator endpoints. Create service groups from the port matrix below.
3. Create the custom zone, if the installed release supports custom zones. Add narrow allows for DHCP to the gateway, assigned DNS, time, identity, updates and any explicitly selected storage service.
4. Add and temporarily log the final inter-zone deny. Put established and related return handling where the installed policy model requires it; do not create a reverse broad allow.
5. Inspect ordering and counters while the zone has no port or SSID. Save screenshots or an exported policy list as the pre-attachment record.
6. Create the virtual network and DHCP scope, but attach no production port or SSID. Assign one sacrificial access port, renew its lease, and run every positive and negative probe.
7. Only after evidence matches the plan, attach the intended port profile or SSID. Rollback means removing that attachment first; policy objects can remain disabled for diagnosis.

Service and port contract

Start with these protocol surfaces, then reduce them from observed service requirements. Stateful return traffic is not a separate client-bound allow. The destination must be an endpoint group, never an entire private root.

Purpose	Source → destination	Transport / port	Verification
DHCP	client → UDM DHCP	UDP 67–68	Capture one discover/offer/request/ack; one server identifier only.
DNS	clients → AD DNS	TCP/UDP 53	Resolve an external name, an AD host and the LDAP/Kerberos SRV records.
Time	clients → approved NTP	UDP 123	Record source, offset and synchronized state.
Kerberos	clients → domain controllers	TCP/UDP 88; TCP/UDP 464	Acquire and inspect a user ticket; test password change separately.
Directory	clients → domain controllers	TCP/UDP 389; TCP 636 if configured; TCP 3268–3269 when used	Bind by DNS name and perform the intended directory lookup.
AD file/RPC	clients → domain controllers	TCP 445, 135; negotiated high TCP RPC range	Join a disposable client and inspect flows before narrowing the configured RPC range.
Boot discovery	build client → UDM and boot host	DHCP plus UDP 69 only if the first stage uses TFTP	Record offered boot server/name and fetch the exact first-stage hash.
Boot payload	build client → boot publication	TCP 80 or 443 as configured	Fetch the immutable manifest and verify every checksum.
Storage	managed clients → named NAS endpoints	TCP 445	Authenticate as a test user; then block storage and prove login remains usable.
Updates	managed clients → Internet	TCP 80/443 via assigned DNS	Complete a metadata refresh and one signed package verification.
Administration	explicit admin sources → exact managed endpoints	HTTPS/SSH or vendor ports actually selected	Connect from an allowed source; repeat from a workstation and observe denial.

RPC is a design choice

Windows domain join and management may negotiate dynamic RPC ports. First measure a disposable join. If a smaller range is required, configure the servers and clients to use that range before narrowing the firewall; simply blocking the default high range produces intermittent, misleading failures.

Executable proof cycle

Run the cycle from a named sacrificial client after each network, policy or service change. Save commands and timestamps in the private evidence ledger.

1. **Link:** record switch port or SSID, client MAC, link speed and observed VLAN. A wrong VLAN stops the cycle.
2. **Lease:** release and renew. Record address/prefix, gateway, DNS, lease server and expiry. Capture DHCP when the authority is ambiguous.
3. **Name and time:** query external A/AAAA and AD SRV records through the assigned resolver. Record the time source and offset.
4. **Required services:** acquire a Kerberos ticket, perform a directory lookup, fetch the boot manifest or update metadata, and open only the selected SMB resource.
5. **Forbidden neighbours:** probe management, another restricted client and one unrelated private zone with DNS, TCP connection and application tests. Record the exact deny rule or traffic-flow entry.
6. **Failure injection:** make storage unavailable and prove login continues; make the boot alias unavailable and prove an installed workstation still boots locally.
7. **Restart:** cold-boot the client and repeat the short lease, DNS, ticket and deny checks. A cached successful session is not cutover evidence.

Right-sized worked example

Suppose a role has six devices today and may reach ten before review. Allocate one /28: 16 total addresses, 14 usable. Use the first usable address as gateway, reserve the next few simple addresses for durable endpoints, and place the dynamic pool above them. Record an adjacent unused block for future growth, but do not create or scan it. Ten devices consume 71 percent of the usable addresses, which triggers review. The private inventory substitutes the exact CIDR and reservations;

the public document never does.

IPv6 phase policy

Phase one validates IPv4 only. This is a scope decision, not permission for unmeasured IPv6. For every new network, record whether the UDM advertises an IPv6 prefix, whether clients receive global or unique-local addresses, and which IPv6 firewall policy applies. Until an IPv6 traffic contract exists, disable prefix delegation and router advertisements on these purpose-built zones where the installed UniFi release permits it, and deny routed IPv6 between trust zones. Link-local IPv6 remains normal on interfaces.

Before enabling routed IPv6, reproduce the complete IPv4 contract with IPv6 DNS answers, explicit service destinations, positive probes, negative probes, logging and rollback. Do not assume an IPv4 deny covers IPv6, and do not call a network isolated while an untested IPv6 route bypasses its IPv4 policy.

PXE safety gate

Do not attach boot options to the ordinary LAN. A dedicated provisioning network makes physical attachment part of erase authorization. Boot options point to a stable service name or reserved address; a later Controller move must not require rebuilding workstations.

AD DNS ordering

Set clients to the AD DNS address only after that resolver answers both `_ldap._tcp.dc._msdcs.<identity-domain>` and external names. AD clients must use AD DNS; a resolver with no working forward path makes the Internet appear broken even when routing is correct.

Managed-workstation Wi-Fi

The managed SSID is post-install connectivity, not the first firmware PXE transport. It maps to the managed-workstation network and has restricted Internet access. The household Wi-Fi secret never enters deployment material.

1. Create the managed-workstation virtual network and prove it on a wired access port first. **Why:** routing and identity failures can then be separated from radio and credential failures.
2. Create a dedicated SSID mapped to that network. Use the UniFi-supported WPA2/PPSK combination when per-device private pre-shared keys are required. **Constraint:** PPSK currently excludes WPA3 and 6 GHz; broadcast this SSID on 2.4 and 5 GHz only.
3. Issue a distinct PPSK per managed workstation and record only its identifier and assignment in the private inventory. Store the secret in the encrypted deployment vault. **What this does:** one lost machine can be revoked without changing every workstation. A copied PPSK still works from another device; AD, not the PPSK, establishes user identity.
4. Inject the credential during an authorised build. The unencrypted phase-one workstation can still reveal its local credential to a physical attacker; segmentation limits what that credential can reach but does not make the disk confidential.
5. Join two test clients. Verify Internet, AD DNS, Kerberos and optional SMB; then verify that management, peer clients and other household zones remain unreachable.

PPSK capability gate and fallback

Before designing around PPSK, confirm that the installed access points, Network release and chosen radio/security mode expose Private Pre-Shared Keys. PPSK may require WPA2 and may be unavailable with WPA3 or 6 GHz operation. If it is unavailable, use a dedicated restricted WPA2 SSID with one deployment credential as the temporary fallback, rotate it after a lost device, and rely on AD for user identity. Prefer 802.1X/RADIUS as the later per-device or per-user design. Never silently place managed clients on household Wi-Fi.

When an isolated SSID appears broken

Symptom	Measure first	Likely boundary
No address	DHCP exchange and switch/AP network assignment	VLAN not carried to the AP, wrong SSID mapping, or DHCP scope absent.
Address but no names	DHCP DNS value; query UDM and AD DNS separately	Firewall omitted DNS or the client received the wrong resolver.
Internet but no domain login	Time offset, AD SRV records, Kerberos and LDAP reachability	AD DNS or clock flow missing; broad Internet access is irrelevant.
Domain login stalls when NAS is down	Block storage explicitly and time login	Home/share mapping is blocking instead of optional.
No peer traffic	Repeat from two clients and inspect isolation setting	Expected on guest-style isolation; disable only if the traffic contract requires peers.
Unexpected private access	Test each zone and inspect the matched rule counter	Broad allow precedes the intended deny or the source network is misclassified.

Migration without an outage

Leave the existing LAN untouched while one new zone is proven. Start with the isolated test network, then identity and boot, storage, and managed workstations. Move one sacrificial client at a time. A zone becomes ordinary only after DHCP, DNS, allowed flows, denied flows, cold reboot and rollback are all observed.

Gate	Pass evidence	Retreat
Network exists	Backup identifier, CIDR calculation and one correct lease	Remove the sacrificial port assignment; delete no old network.
Core flows	DNS SRV, clock, Kerberos and boot fetch pass by stable names	Disable new rules and return the test port to its recorded old profile.
Denies work	Forbidden probes fail and hit the intended counters	Disable the final deny, repair missing specific allows, repeat.
Wi-Fi works	Two clients prove allowed and forbidden paths	Disable the new SSID; wired proof remains available.
Client moved	Login passes online and with optional storage unavailable	Restore its former network assignment; no workstation rebuild.

Future placement register

Network names and service discovery deliberately outlive physical placement. These roles may later move from a host into VMs without workstation rebuilds: AD and integrated DNS, boot publication, deployment credential broker, certificate authority, Windows deployment helpers, and audit services. Move stateful identity by replication and role transfer; move boot service by its stable alias and UniFi reservation. Never rename the realm to describe a host.

Measurements before policy

A firewall rule is not finished when it saves. Measure the path from a named source client, record the destination address actually selected by DNS, and match the observation to a UniFi flow or counter.

Layer	Question	Measurement	Failure means
Link	Is the client on the intended access network?	Port/SSID assignment, client MAC and observed VLAN	Repair carriage; do not touch firewall rules.
Address	Did the sole authority answer?	Lease, CIDR, gateway, DNS and DHCP server identifier	Repair scope or guarding; do not add a static client address.
Name	Did AD DNS choose the intended service?	A/AAAA answer and AD SRV answer from the assigned resolver	Repair DNS or forwarding; an IP-only probe is insufficient.
Route	Did the packet cross the intended zones?	Source/destination addresses and matched UniFi flow	Repair zone membership or rule order.
Service	Did the complete protocol succeed?	Kerberos ticket, SMB session, HTTP checksum or update transaction	Inspect the named dependency; do not create <i>allow any</i> .
Failure path	Does the promised deny really hold?	Forbidden probe plus blocked-flow evidence	A silent timeout alone does not identify the enforcing boundary.

Rule record

Every allow rule records: purpose, owner, source zone and object, destination name and object, protocol/service group, whether return traffic is automatic, the successful probe, a forbidden neighbouring probe, creation date and review date. Temporary rules carry an expiry. A broad diagnostic rule may exist only while attended and is removed before the test is called complete.

Official operating references

These sources describe the current feature contracts. Record the installed UniFi Network version because exact menu labels may change.

- Creating UniFi virtual networks and VLANs
- UniFi DHCP server and Network Boot options
- Zone-based firewall ordering and Gateway-zone warnings
- Duplicate addressing and DHCP Guarding
- PPSK VLAN mapping and radio-security limits
- Network, client and device isolation layers
- Traffic-flow inspection and logging

Final network worksheet

Keep the completed copy in the private repository.

Record	Instance value or evidence
UniFi OS / Network versions	_____
Backup identifier and restore test	_____
Site allocation root	_____
Zone, VLAN, CIDR and gateway	_____
Expected / maximum population	_____
DHCP pool and reservations	_____
DNS and time sources	_____
Required allow-rule evidence	_____
Forbidden-flow evidence	_____
PXE server and boot filename	_____
SSID/PPSK assignment reference	_____
Rollback observation	_____

Stop condition

If the observed source network, DHCP authority, DNS answer, rule counter or target service does not match the frozen plan, stop. Restore the test port or disable the new SSID. Do not widen a rule until the missing dependency has a name, a narrow flow and a repeatable proof.

Cutover evidence ledger

Run the same probes before and after the change. Name the saved capture, log export or screenshot; “worked” is not a record.

Time	Probe and source zone	Pass / fail	Evidence file or exact observation
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Deviation or unresolved question:

Operator: _____ Date: _____ Review due: _____

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